

Gelei Deng

+65-8403-0341, gelei.deng@ntu.edu.sg

Profile

Ph.D. in Computer Science specializing in Cybersecurity and AI Safety, with a track record of translating research into widely adopted, real-world security systems. Creator of PentestGPT, a 10k+ star open-source LLM-powered penetration testing framework with active global contributors. Extensive experience in frontier model red teaming, vulnerability discovery, and security automation, including work with OpenAI's Red Teaming Network and multiple Anthropic Model Safety Bug Bounty awards. Proven ability to operate across academia, industry, and adversarial settings to secure deployed AI systems at scale.

Areas of Expertise

- AI Security & Model Red Teaming (model jailbreaking, prompt injection, multimodal attacks)
- Security Automation with LLMs (agentic testing, static + dynamic analysis, exploit discovery)
- Software and Smart Contract Security
- Applied Security Research & Open-Source Systems Development and Management
- Programming: Python, Solidity; experience with security tooling across major platforms

Education

Nanyang Technological University (Aug 2020 – Oct 2024)

- Ph.D. Computer Science, Cybersecurity
- NTU Cyber Security Lab
- Main Research Interests on System Security, Web Security and Penetration Testing

Singapore University of Technology and Design (May 2015 – Sep 2018)

- Department of Engineering Product Design, B.E. Electrical Engineering, Honor List
- Singapore Ministry of Education Full Scholarship (SM2) Holder

Employment

Nanyang Technological University (Dec 2024 - Present)

Research Assistant Professor

- Conduct pioneering research on AI security, with focus on AI safety, LLM Red Teaming, and AI for Security Automation at NTU Digital Trust Center.
- Lead academic initiatives on LLM attacks and mitigation techniques for deployed AI systems.
- Supervise Ph.D. students and design graduate-level courses in cybersecurity and AI security.
- Translate research into open-source tools and industry collaborations, bridging academic insights with real-world security deployment.

OpenAI (Jan 2024 – Present, part-time)

Independent Contractor at OpenAI Red Teaming Network

- Participate in OpenAI-led red teaming to assess risks and safety profiles of OpenAI models and systems (e.g., GPT4o, DALL-E3, Operator, GPT5).
- Identify, document, and report high-impact safety and security vulnerabilities, contributing to model risk assessment and mitigation.

Quantstamp, Inc. (Oct 2022 – Present)

Lead AI Research Engineer, Auditing Engineer

- Conduct comprehensive security audits for blockchain projects (DeFi, wallets, DEXs, NFTs).
- Lead R&D on automated on-chain vulnerability identification with LLM-enhanced static analysis, fuzzing and exploit generation.

Projects and Awards

PentestGPT: An LLM-empowered Automatic Penetration Testing Tool

- The [world-first LLM-enhanced penetration testing agentic framework](#).
- The research paper accepted by USENIX Security 2024 with [distinguished artifact award](#), current citation 300+.
- Open-source tool with **10k+ stars** on GitHub: <http://pentestgpt.com/>, with active community and world-wide contributors.
- Various collaboration with industrial partners (TikTok, Huawei, Darktrace, etc.)

Bug Bounties, CVEs and Vulnerability Identification

- Submitted multiple high-impact vulnerability reports via OpenAI Red Team Network, focusing on jailbreaking and prompt-injection attacks against LLM agents (e.g., OpenAI Operator).
- Two-time winner of the [Anthropic Model Safety Bug Bounty](#), earning \$20K+ USD in total.
- Identified and disclosed 10+ verified CVEs, including CVE-2021-39114, CVE-2021-30224, CVE-2021-37392, CVE-2021-37393, CVE-2021-37476, and CVE-2021-37477.
- Discovered and responsibly disclosed vulnerabilities confirmed by major vendors, including Atlassian Confluence, Apache Magento, Bitbucket, SEO Panel, and Spree Commerce.

Publication

- [3,700+ citations since 2021](#) · h-index: 22 · i10-index: 34
- A complete list of publication: <https://scholar.google.com/citations?user=NuzUkMEAAAAJ>

Highlighted Research Work

Gelei Deng, Haoran Ou, Yi Liu, Jie Zhang, Tianwei Zhang, Yang Liu, “Oedipus: LLM-enhanced Reasoning CAPTCHA Solver,” in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (ACM CCS)*, 2025.

Ziqi Ding, **Gelei Deng***, Yi Liu, Junchen Ding, Jieshan Chen, Yulei Sui, and Yuekang Li, “IllusionCAPTCHA: A CAPTCHA based on Visual Illusion,” In *Proceedings of the ACM on Web Conference 2025 (WWW '25)*. Association for Computing Machinery, New York, NY, USA.

Cheng Wang, **Gelei Deng***, Xianglin Yang, Han Qiu, Tianwei Zhang, “When Audio and Text Disagree: Revealing Text Bias in Large Audio-Language Models,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025.

Chenhang Cui, **Gelei Deng***, An Zhang, Jingnan Zheng, Yicong Li, Lianli Gao, Tianwei Zhang, Tat-Seng Chua, “Safe + Safe = Unsafe? Exploring How Safe Images Can Be Exploited to Jailbreak Large Vision-Language Models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

Gelei Deng, Yi Liu, Víctor Mayoral-Vilches, Peng Liu, Yuekang Li, Yuan Xu, Tianwei Zhang, Yang Liu, Martin Pinzger, Stefan Rass, “PentestGPT: An LLM-empowered Automatic Penetration Testing Tool,” in *33rd USENIX Security Symposium (USENIX '24)*. [Distinguished Artifact Award](#), 2024.

Gelei Deng, Yi Liu, Kailong Wang, Yuekang Li, Tianwei Zhang, Yang Liu, “PANDORA: Jailbreak GPTs by Retrieval Augmented Generation Poisoning,” in *Workshop on Artificial Intelligence System with Confidential Computing (AISCC)*, [Distinguished Paper Award](#), February, 2024.

Gelei Deng, Yi Liu, Yuekang Li, Kailong Wang, Ying Zhang, Zefeng Li, Haoyu Wang, Tianwei Zhang, Yang Liu, “MASTERKEY: Automated Jailbreaking of Large Language Model Chatbots,” in *Network and Distributed System Security Symposium (NDSS '24)*. 2024.

Gelei Deng, Zhiyi Zhang, Yuekang Li, Yi Liu, Tianwei Zhang, Yang Liu, Guo Yu, Dongjin Wang, “NAUTILUS: Automated RESTful API Vulnerability Detection,” in *32nd USENIX Security Symposium (USENIX '23)*. 2023.

Gelei Deng, Guowen Xu, Yuan Zhou, Tianwei Zhang, and Yang Liu, “On the (In)Security of Secure ROS2,” in *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security (CCS '22)*. 2022

Gelei Deng, Yuan Zhou, Yuan Xu, Tianwei Zhang, and Yang Liu, “An Investigation of Byzantine Threats in Multi-Robot Systems,” in *24th International Symposium on Research in Attacks, Intrusions and Defenses (RAID '21)*. 2021